

Resource Management System

Pseudocode

BEGIN

Declare Resource array R[100]

Set count = 0

Load previously stored records from file

DO

Display Main Menu

1. Add New Resource
2. Update Resource
3. Display All Resources
4. Search Resource
5. Sort Resources
6. Merge Department Records
7. Identify Duplicate Resources
8. Analyse Resource Availability
9. Display Critical / Low Resources
10. Generate Consolidated Report
11. Save Records
12. Load Records
13. Exit

Read choice

SWITCH choice

CASE 1:

Add new resource

Read resource ID, name, category,
department, unit, quantity,
minimum stock level and priority

Check for duplicate ID

IF duplicate ID exists THEN

Display "Duplicate Resource ID"

ELSE

Store resource in array

Increment count

Display "Resource added successfully"

END IF

BREAK

CASE 2:

Read Resource ID

Search for the resource

IF resource is found THEN

Read new quantity

Read new minimum stock level

Read new priority

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        Read new department

        Update resource details

        Display "Resource updated successfully"
    ELSE
        Display "Resource not found"
    END IF
    BREAK

CASE 3:
    IF count = 0 THEN
        Display "No resource records available"
    ELSE
        FOR i = 0 TO count - 1
            Display all details of R[i]
        END FOR
    END IF
    BREAK

CASE 4:
    Display Search Menu

    1. Search by ID
    2. Search by Name
    3. Search by Category
    4. Search by Department

    Read search choice

    Search the array using the selected criterion

    IF resource is found THEN
        Display resource details
    ELSE
        Display "Resource not found"
    END IF
    BREAK

CASE 5:
    Display Sorting Menu

    1. Sort by Quantity
    2. Sort by Priority
    3. Sort by Department
    4. Sort by Resource Name

    Read sort choice

    Sort the resource array using the selected criterion

    Display sorted resources
    BREAK

CASE 6:
    Read resource records from another department

    FOR each new resource
        Compare its ID with existing records
        IF matching ID is found THEN

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        Add the new quantity to existing quantity
        Display "Duplicate resource merged"
    ELSE
        Add new resource to the array
        Increment count
    END IF
END FOR

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Display "Department records merged successfully"
BREAK

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CASE 7:

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Set duplicateFound = FALSE

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FOR i = 0 TO count - 2
    FOR j = i + 1 TO count - 1

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        IF R[i].ID = R[j].ID THEN
            Display duplicate resource
            Set duplicateFound = TRUE
        END IF

```

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    END FOR
END FOR

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IF duplicateFound = FALSE THEN
    Display "No duplicate resources found"
END IF
BREAK

```

CASE 8:

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FOR i = 0 TO count - 1

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    Calculate critical level
    critical level = 25% of minimum stock level

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    IF quantity < critical level THEN
        Status = "CRITICAL"

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    ELSE IF quantity < minimum stock level THEN
        Status = "LOW"

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    ELSE
        Status = "ADEQUATE"
    END IF

```

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    Display resource name and status

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END FOR
BREAK

```

CASE 9:

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Display Critical Resources

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FOR i = 0 TO count - 1
    IF quantity < 25% of minimum level THEN
        Display resource
    END IF

```

```

END FOR
Display Low Stock Resources

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```

FOR i = 0 TO count - 1
    IF quantity < minimum level AND
        quantity >= 25% of minimum level THEN
        Display resource
    END IF
END FOR
BREAK

```

CASE 10:

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Set totalQuantity = 0
Set adequateCount = 0
Set lowCount = 0
Set criticalCount = 0

FOR i = 0 TO count - 1

    totalQuantity =
    totalQuantity + R[i].quantity

    IF quantity < 25% of minimum level THEN
        Increment criticalCount

    ELSE IF quantity < minimum level THEN
        Increment lowCount

    ELSE
        Increment adequateCount
    END IF

END FOR

Display total resources
Display total available quantity
Display adequate resources
Display low-stock resources
Display critical resources
Display department-wise availability
Display resources requiring immediate replenishment
BREAK

```

CASE 11:

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Open "resources.dat" in write mode

IF file cannot be opened THEN
    Display "File opening error"
ELSE
    Write count to file

    FOR i = 0 TO count - 1
        Write R[i] to file
    END FOR

    Close file
    Display "Records saved successfully"
END IF
BREAK

```

CASE 12:

```

Open "resources.dat" in read mode
IF file does not exist THEN

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        Display "No previous records found"
    ELSE
        Read count from file

        FOR i = 0 TO count - 1
            Read R[i] from file
        END FOR

        Close file
        Display "Records loaded successfully"
    END IF
    BREAK

CASE 13:
    Save records to file
    Display "Program terminated"
    BREAK

DEFAULT:
    Display "Invalid choice"

END SWITCH

WHILE choice != 13

FUNCTION CountCritical(index)

    IF index = count THEN
        RETURN 0
    END IF

    IF R[index].quantity <
        25% of R[index].minimumLevel THEN

        RETURN 1 + CountCritical(index + 1)

    ELSE

        RETURN CountCritical(index + 1)

    END IF

END FUNCTION

END

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